Confidence-Aware Network Digital Twins for Safe Autoscaling of 5G Core Services

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***Abstract*—Network Digital Twins can support data-driven network management by providing a controllable environment for policy training and what-if analysis. Their predictions, however, may become inaccurate because of incomplete telemetry, measurement delay, workload changes, resource interference, or configuration drift. This issue is particularly important when a Digital Twin is used to train reinforcement-learning policies for autoscaling critical network services. This paper proposes a confidence-aware Network Digital Twin for safe autoscaling of 5G core services. The approach associates each predicted state transition with an uncertainty estimate. Depending on this confidence, the controller selects a Digital-Twin-based action, a guarded action, or a conservative fallback policy. The proposal combines state synchronization, functional service representations, uncertainty-aware prediction, and constrained reinforcement learning. An evaluation methodology based on a containerized core-network testbed is defined. The objective is to make Digital-Twin-based closed loops more robust, auditable, and suitable for operational service management.**

***Index Terms*—Network Digital Twin, 5G core, autoscaling, safe reinforcement learning, service assurance, uncertainty estimation.**

I. Introduction

Telecommunication networks are increasingly implemented as cloud-native infrastructures composed of virtual machines, containers, and microservices. This evolution facilitates flexible deployment and resource sharing, but it also increases the complexity of management. The capacity required by a network service varies with user mobility, application demand, special events, and failures. Autoscaling must therefore adapt resources while preserving service-level objectives.

In a packet core, a scale-out action can prevent overload but increases resource consumption. A scale-in action reduces cost but may cause session loss or service degradation if the remaining capacity is insufficient. Reinforcement learning (RL) is attractive for this problem because it can learn policies that balance several objectives. Deep RL has been applied to packet-core autoscaling [1], and state prediction has been integrated to anticipate network evolution [2]. Safe RL has subsequently been investigated to constrain unsafe exploration and improve the applicability of learned policies [3].

A complementary challenge concerns the environment used by the learning algorithm. A Network Digital Twin (NDT) can provide a controllable environment in which management policies are evaluated without affecting the operational network. However, an NDT is not an exact replica of the physical system. It can become inaccurate when telemetry is delayed, workload distributions change, resources are contended, or the software configuration evolves.

This paper proposes a confidence-aware NDT for safe closed-loop autoscaling of 5G core services. The central idea is to associate each predicted state transition and candidate action with a confidence estimate. The controller uses this estimate to determine whether an action can be selected from the NDT prediction, must be checked by a safety guard, or should be replaced by a conservative fallback action.

The contributions are:

- an architecture for confidence-aware NDT-based autoscaling;
- a state representation combining infrastructure, workload, service, and QoE information;
- a decision mechanism combining uncertainty estimation, constrained RL, and fallback control;
- an evaluation methodology covering fidelity, safety, resource efficiency, and robustness to drift.

The proposal is consistent with previous work by the author on packet-core autoscaling, state prediction, Safe RL, QoE prediction, and network configuration management. No experimental results are claimed; the paper presents an implementable design and evaluation protocol.

II. Background and Problem Formulation

A. Digital Twins for Network Management

A Digital Twin is a software representation of a physical or operational system that is updated using observations from that system. In network management, it may include topology, configuration, resource allocation, workload, performance, and failure information. Digital Twins can support monitoring, planning, what-if analysis, anomaly diagnosis, and autonomous control [7].

For autoscaling, the twin should reproduce the relationship between workload, allocated capacity, and service performance. Relevant variables include connected users, attach rate, session-establishment rate, CPU utilization, memory usage, instance count, dropped sessions, latency, and availability.

A simulator generates behavior from an abstract model. An NDT must additionally maintain a relationship with an operational system through telemetry, synchronization, and model updating. This distinction is important for closed-loop

control. A model that is accurate during training may become unsuitable after a workload or configuration change. Moreover, prediction error close to a service threshold is more critical than the same error during a low-load period.

B. Constrained Autoscaling

Let s_t be the state of a service at time t. The action set is

$$\mathcal{A} = \{-1, 0, +1\}, \tag{1}$$

where the actions represent scale-in, no change, and scale-out, respectively. An RL policy π maximizes a discounted reward

$$J_R(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T} \gamma^t r(s_t, a_t) \right], \tag{2}$$

where the reward can combine resource cost, scaling operations, latency, throughput, and availability.

A separate safety cost is defined as

$$J_C(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T} \gamma^t c(s_t, a_t) \right]. \tag{3}$$

The control problem is

$$\max_\pi J_R(\pi) \quad \text{subject to} \quad J_C(\pi) \le C_{\max}. \tag{4}$$

The safety cost may include dropped sessions, latency violations, service unavailability, or an insufficient capacity margin. Separating reward and cost prevents a small resource saving from compensating for a severe service violation.

C. Service and QoE Information

Infrastructure metrics do not always represent the experience of end users. Previous work by the author compared functional data analysis and signature-based representations for predicting Voice over IP QoE from high-frequency measurements [4]. These measurements are naturally represented as trajectories rather than as independent scalar samples.

Accordingly, the proposed state includes a recent measurement window. Functional basis coefficients, truncated signatures, or a fixed history of samples can be used to represent this window. Functional mixture-of-experts models are also relevant when service behavior contains several regimes, such as low load, congestion, and recovery after scaling [5].

III. Confidence-Aware Digital Twin

A. Architecture

The proposed NDT contains four logical components:

1) a state synchronizer;
2) a predictive service model;
3) a confidence estimator;
4) a what-if execution interface.

The state synchronizer aligns telemetry collected at different frequencies, handles missing observations, records measurement age, and associates each state with the active configuration.

The predictive model estimates service behavior after a candidate action. It may be data-driven, analytical, or hybrid. It need not reproduce every internal detail of the network; it must predict the variables used by the management policy and safety guard.

The confidence estimator evaluates whether a state and candidate action are within the validated domain of the twin. It uses prediction uncertainty, distance from the training distribution, and recent prediction residuals.

The what-if interface evaluates candidate actions without modifying the physical network. For a state s_t and action a, it returns

$$\mathcal{T}(s_t, a) = (\hat{s}_{t+1}, \hat{r}_{t+1}, \hat{c}_{t+1}, \boldsymbol{\sigma}_{t+1}), \tag{5}$$

where $\boldsymbol{\sigma}_{t+1}$ contains uncertainty estimates.

B. State Representation

The controller state is defined as

$$s_t = [x_t, u_t, q_t, h_t, m_t], \tag{6}$$

where x_t contains infrastructure variables, u_t workload variables, q_t service and QoE variables, h_t recent history, and m_t deployment and configuration information.

Infrastructure variables include CPU, memory, bandwidth, instance count, and placement. Workload variables include active users, attach rate, session-establishment rate, and traffic volume. Service variables include latency, dropped sessions, throughput, and availability. The history enables the controller to distinguish increasing, decreasing, and stable utilization. Configuration variables include software version, resource limits, scaling bounds, and deployment topology.

Telemetry age is explicitly included. A value collected recently must not be treated as equivalent to a value collected several control intervals earlier.

C. Uncertainty Estimation

Let f_k be the kth model in an ensemble of K predictive models. For a state and action, the models produce

$$\hat{s}_{t+1}^{(k)} = f_k(s_t, a), \qquad k = 1, \dots, K. \tag{7}$$

The ensemble mean is

$$\bar{s}_{t+1} = \frac{1}{K} \sum_{k=1}^{K} \hat{s}_{t+1}^{(k)}. \tag{8}$$

For a critical metric y, ensemble disagreement is estimated by

$$\sigma_y^2 = \frac{1}{K} \sum_{k=1}^{K} \left(\hat{y}_{t+1}^{(k)} - \bar{y}_{t+1} \right)^2. \tag{9}$$

This estimates epistemic uncertainty, that is, uncertainty caused by limited knowledge of the system. It does not completely represent randomness inherent in traffic or service

behavior. Prediction intervals can be calibrated using held-out data or conformal prediction [9].

Let $\mathcal{Y}_c$ be the set of critical service metrics. The confidence score for candidate action a is

$$\kappa_t(a) = 1 - \text{clip}\left(\sum_{y \in \mathcal{Y}_c} w_y \tilde{\sigma}_y + w_d d(s_t) + w_r e_t, 0, 1\right), \tag{10}$$

where $\tilde{\sigma}_y$ is normalized uncertainty, $d(s_t)$ is the distance from the training distribution, and e_t is a recent prediction residual. The residual can be computed as

$$e_t = \|s_t - \hat{s}_t\|_{\mathbf{W}}, \tag{11}$$

with larger weights assigned to service variables.

The confidence score is an operational indicator, not a probability of safety. A high score means that the twin has behaved reliably for similar states and actions; it does not guarantee that a violation cannot occur.

IV. Closed-Loop Controller

A. Operating Modes

The controller operates in three modes.

Twin mode is used when confidence is high. Candidate actions are evaluated using the NDT, and the action with the highest predicted objective is selected subject to safety constraints.

Guarded mode is used when confidence is intermediate. An action is accepted only if its uncertainty interval satisfies the service constraints. The controller may restrict the action set by disabling scale-in and allowing only scale-out or no change.

Fallback mode is used when confidence is low, telemetry is unavailable, or persistent prediction errors are detected. A deterministic policy is then applied. For example, the system can scale out at a conservative workload threshold and scale in only after a long stable period.

Let κ_H and κ_L denote high and low confidence thresholds. Hysteresis is used to avoid frequent mode changes. The controller must remain above κ_H for several intervals before returning to twin mode and must remain in fallback mode for a minimum duration after a safety event.

B. Safety Guard

For each candidate action, the NDT predicts intervals for critical metrics. The safe action set is

$$\mathcal{A}_{\text{safe}}(s_t) = \{a \in \mathcal{A} : U_y(s_t, a) \le y_{\max},\ \forall y \in \mathcal{Y}_{\text{upper}}\}, \tag{12}$$

where U_y is the upper prediction bound for a metric with an upper limit. Lower bounds are used for metrics with minimum requirements.

The selected action is

$$a_t = \arg\max_{a \in \mathcal{A}_{\text{safe}}(s_t)} \left[\hat{J}_R(s_t, a) - \lambda(1 - \kappa_t(a))\right]. \tag{13}$$

The uncertainty penalty discourages actions whose outcomes are poorly known. If the safe action set is empty, the fallback policy is selected. This rule is intentionally conservative: uncertainty is not used as an opportunity for exploration in a critical service.

C. Training and Transfer

The policy is trained in the NDT using both reward and safety cost. PPO and constrained variants are possible choices because they support discrete actions and constrained policy updates [8]. During training, workload, resource capacity, measurement delay, and model parameters can be perturbed to reduce overfitting.

Before operational use, the policy is evaluated in the NDT on held-out workloads and then transferred through a guarded deployment process. During execution, predicted and observed outcomes are compared. A large residual reduces confidence and can trigger fallback operation.

V. Synchronization and Model Updating

After an action is executed in the physical network, its observed outcome is associated with the corresponding state, action, configuration, and workload. This observation is used to calculate the prediction residual and update the NDT dataset.

A model update is considered when:

- prediction residuals exceed a predefined threshold;
- errors increase in a particular workload regime;
- the software or infrastructure configuration changes;
- an out-of-distribution detector reports a persistent shift.

A candidate model should first be evaluated offline. Average prediction error is insufficient: the model must also be accurate near safety boundaries and provide calibrated confidence estimates. A shadow-update process can train a new model while the current model remains active. Promotion is performed only after the candidate satisfies minimum fidelity and safety requirements.

Configuration must be treated as part of the twin state. Changes to a Helm value, network-function version, resource limit, or placement policy can modify service behavior. Versioned configuration information is therefore needed for reproducibility and drift diagnosis.

VI. Evaluation Methodology

A. Research Questions

The evaluation addresses four questions:

RQ1: Does low confidence identify states with higher NDT prediction error?

RQ2: Does confidence-aware selection reduce service violations compared with an NDT controller without confidence?

RQ3: What resource-efficiency cost is introduced by guarded and fallback operation?

RQ4: How quickly does synchronization recover NDT quality after workload or configuration drift?

B. Platform and Baselines

The experiments can use a containerized packet-core platform deployed on Kubernetes. The platform should expose metrics from network functions, containers, and compute nodes. The initial implementation can use Magma, which was used in previous Safe RL experiments [3]. Open5GS or another cloud-native core can be used for additional validation.

The comparison should include:

1) threshold-based autoscaling;
2) unconstrained deep RL;
3) Safe RL without an NDT;
4) Safe RL using an NDT without confidence estimation;
5) the proposed confidence-aware controller.

These baselines isolate the effects of RL, safety constraints, Digital-Twin prediction, and confidence-aware control.

C. Scenarios and Metrics

Training and test data should be separated temporally. The test set should include at least one workload profile and one configuration condition not used during training. Scenarios should include stationary low and high load, gradual increase, sudden bursts, rapid recovery, telemetry delay, missing measurements, CPU contention, node capacity limits, and network-function restart.

The main metrics are grouped as follows:

- **Safety**: dropped sessions, latency violations, service unavailability, and violation duration.
- **Efficiency**: instance count, CPU and memory use, scaling operations, and time above required capacity.
- **Twin fidelity**: one-step and multi-step prediction error, interval coverage, calibration error, and confidence-error correlation.
- **Robustness**: fallback frequency, fallback duration, recovery time after drift, and rejected actions.

Average values should be complemented by worst-case indicators, such as maximum violation duration and high-percentile latency. Multiple random seeds should be used for RL training.

An ablation study can remove history information, QoE variables, configuration identifiers, residual monitoring, or hysteresis. This determines which components provide the largest benefit.

VII. Discussion

When the workload remains within the training distribution, the confidence score should remain high and the controller should operate mostly in twin mode. Predictive scaling can then reduce the need for reactive overprovisioning.

For an unfamiliar traffic burst, ensemble disagreement and out-of-distribution distance should increase. The controller should enter guarded or fallback mode and prefer a conservative scale-out decision. This may consume more resources than an unconstrained controller, but it reduces the risk of selecting a scale-in action based on an inaccurate prediction.

The principal trade-off is between safety and efficiency. A low fallback threshold may preserve resources but tolerate more model uncertainty. A high threshold provides stronger protection but may increase conservative operation. This trade-off should be reported using safety-efficiency curves rather than a single aggregate score.

The approach also supports operational auditability. For each action, the system can record the synchronized state, candidate actions, twin predictions, confidence score, safety constraints, selected mode, and observed outcome. Such information can support incident analysis and model improvement.

The proposal is compatible with Network-as-Code practices in which network configuration is versioned and deployed through automated workflows. Previous work on generating commit messages for 5G configuration files highlights the importance of traceability and contextual information in network configuration management [6].

VIII. Limitations

Ensemble disagreement does not capture all uncertainty. If all models share the same structural bias, they may agree on an incorrect prediction. Residual monitoring and validation on fresh observations are therefore necessary.

The confidence score is application dependent. The weights associated with latency, session drops, availability, and resource utilization must be calibrated for each service. In addition, long action sequences can accumulate prediction errors even when one-step predictions are accurate.

A testbed cannot reproduce every property of a production network. Results obtained with a particular core implementation or Kubernetes cluster should therefore be interpreted as evidence about the methodology rather than as a guarantee for every deployment.

Finally, fallback policies require independent validation. A confidence-aware controller reduces dependence on the twin, but it does not eliminate the need for operational thresholds, capacity margins, and service runbooks.

IX. Conclusion

This paper proposed a confidence-aware Network Digital Twin for safe closed-loop autoscaling of 5G core services. The approach combines state synchronization, functional service representations, predictive modeling, uncertainty estimation, constrained RL, and deterministic fallback control.

The main principle is that a Digital Twin should not be considered an always-correct substitute for the operational network. Its predictions should be accompanied by a confidence estimate that is continuously checked against observations. High-confidence situations can use predictive RL, intermediate situations can use guarded decisions, and low-confidence situations can use conservative fallback policies.

The proposed evaluation methodology measures service safety, resource efficiency, twin fidelity, calibration, and recovery after model drift. Future work will implement the architecture in a CoreNetTwin testbed, evaluate calibrated uncertainty estimators, and study the effects of configuration changes and telemetry delay. Multi-service twins will also be investigated, since scaling one network function may affect the performance and confidence of other services.

REFERENCES

[1] J. Singh, S. Verma, Y. Matsuo, F. Fossati, and G. Fraysse, "Autoscaling packet core network functions with deep reinforcement learning," in *Proc. IEEE/IFIP Network Operations and Management Symposium (NOMS)*, Miami, FL, USA, 2023, pp. 1–6, doi: 10.1109/NOMS56928.2023.10154312.

[2] Y. Matsuo, J. Singh, S. Verma, and G. Fraysse, "Integrating state prediction into the deep reinforcement learning for the autoscaling of core network functions," in *Proc. IEEE/IFIP Network Operations and Management Symposium (NOMS)*, Miami, FL, USA, 2023, pp. 1–5, doi: 10.1109/NOMS56928.2023.10154301.

[3] X. Long and G. Fraysse, "Safe RL for core network autoscaling," in *Proc. 4th International Workshop on Analytics for Service and Application Management (AnServApp)*, 2024.

[4] J. S. Tamo Tchomgui, J. Jacques, S. Chrétien, G. Fraysse, and V. Barriac, "Prédiction de la qualité d'expérience dans les réseaux mobiles : cas de la VoIP," in *Proc. 53es Journées de la Statistique de la Société Française de Statistique*, Lyon, France, 2022, HAL Id: hal-03657249.

[5] J. S. Tamo Tchomgui, J. Jacques, S. Chrétien, G. Fraysse, and V. Barriac, "Modèle de mélanges d'experts pour données fonctionnelles," in *Proc. 54es Journées de la Statistique de la Société Française de Statistique*, Brussels, Belgium, 2023, HAL Id: hal-04087177.

[6] B. Yang, A. Samba, G. Fraysse, and S. Cherrared, "Generating commit messages for configuration files in 5G network deployment using LLMs," in *Proc. 4th International Workshop on Analytics for Service and Application Management*, 2024.

[7] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108952–108971, 2020, doi: 10.1109/ACCESS.2020.2998358.

[8] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[9] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023, doi: 10.1561/2200000101.

APPENDIX A
LLM AND GENERATIVE-AI DISCLOSURE

This appendix documents the use of generative artificial intelligence in the preparation of this manuscript. The disclosure is provided to support transparency and reproducibility. The final responsibility for the scientific content, claims, references, and compliance with the conference requirements remains with the named author.

A. Large Language Model Used

The manuscript was generated with OpenAI ChatGPT accessed through an API-based conversational interface. The exact model identifier, sampling parameters, system configuration, and serving infrastructure were not exposed to the author through the interface used for this interaction. Consequently, this disclosure does not claim a model version, temperature, top-p value, random seed, context-window configuration, or hardware configuration that cannot be independently verified.

No additional language model, fine-tuned model, retrieval-augmented generation system, or external autonomous agent was deliberately configured by the author for the preparation of this manuscript.

B. Initial Prompting Strategy

The initial user instruction requested a five-page CNSM short paper on "Digital Twins for Networks and Services." It specified the following requirements:

- identify an original topic consistent with the state of the art;
- use the LaTeX template required by the CNSM 2026 Call for Papers;
- align the contribution with the previous work of Guillaume Fraysse from Orange Research;
- use the author affiliation "Orange Research, Châtillon, France";
- include no figures;
- verify the consistency and existence of references;
- expand the document base with cited, citing, and relevant reference literature.

The initial strategy was therefore to identify a research problem at the intersection of Network Digital Twins, 5G core-network management, autoscaling, reinforcement learning, service assurance, and uncertainty estimation. The resulting topic was confidence-aware Digital-Twin-based autoscaling of 5G core services.

The following user-visible prompts were subsequently used in the conversation:

1) "generate a paper for this conference"
2) "the paper is only 3-page long, expand it to 5 pages"
3) "please provide the full content in a single answer"
4) "the paper is now 7 pages, the limit is at 5, please provide an updated version with 5 pages"
5) "add an appendix for the disclosure agreement detailing the LLM(s) used, the exact initial prompting strategy, the framework used for code/experiment generation (if any), and the constraints applied. It must include the prompts use before this one and this one and the configuration of the agent."

The first instruction was the main content-generation prompt. The following prompts were revision prompts requesting changes in length, format, completeness, pagination, and disclosure. No hidden user-authored prompt was used in addition to the instructions listed above.

C. Prompting and Revision Process

The manuscript was produced through the following sequence:

1) selection of a specific research topic related to Digital Twins and 5G core autoscaling;
2) drafting of an IEEE-style short paper;
3) expansion of the paper after an initial pagination estimate;
4) consolidation of the complete LaTeX source into one response;
5) condensation after the expanded version exceeded the five-page limit;

6) addition of this disclosure appendix.

The paper was generated without requesting fabricated experimental results. The final version consequently presents an architecture, mathematical formulation, and evaluation methodology rather than claiming experimental measurements that were not supplied.

D. Framework for Code and Experiment Generation

No software framework was used to generate, execute, or validate code or experiments. In particular:

- no Python, MATLAB, R, or other experiment-generating code was executed;
- no simulator or testbed was executed;
- no Kubernetes, Magma, Open5GS, Gym, or reinforcement-learning experiment was run during this interaction;
- no numerical result, graph, table, or performance value was generated from a new experiment;
- no automated LaTeX compiler was used to verify the exact page count.

The LaTeX source was generated as text. The references to Kubernetes, Magma, Open5GS, Gym, PPO, and related technologies describe a proposed implementation and evaluation methodology or refer to prior work; they do not indicate that these tools were executed for this manuscript.

E. Constraints Applied

The following constraints were applied during generation and revision:

- the manuscript had to address "Digital Twins for Networks and Services";
- the paper had to be suitable for the CNSM short-paper format;
- the target length was five pages in the official two-column conference template;
- figures were excluded because they could not be generated in this interaction;
- the author had to be Guillaume Fraysse from Orange Research, Châtillon, France;
- the writing had to remain neutral, scientific, polite, and non-accusatory;
- the contribution had to be consistent with the author's previous work on QoE prediction, packet-core autoscaling, state prediction, Safe RL, and network configuration management;
- claims had to be distinguished from proposed future work;
- unsupported experimental results had to be avoided;
- the bibliography had to use recognized conferences, journals, or clearly identified scholarly repositories where applicable;
- the final source had to be provided as a single LaTeX document.

The five-page requirement is a target rather than a guarantee because pagination depends on the exact CNSM template version, bibliography layout, font substitutions, compiler, and metadata. The source should be compiled with the official conference template before submission.

F. Agent Configuration

The interaction used a conversational assistant configured by the platform with general instructions to act as a research assistant in Network Management. The assistant was additionally instructed to write in a scientific style, use the conference format, align the paper with the specified author, and respect the constraints stated in the user request.

The platform-level system instructions and developer instructions are not user-visible configuration data and are not reproduced here. Accordingly, this disclosure reports the functional configuration relevant to the manuscript but does not claim to provide the exact hidden system prompt, developer prompt, internal tool configuration, model routing configuration, or safety configuration.

The assistant had access to a documentary knowledge-base search tool. The tool was used to retrieve information about publications associated with Guillaume Fraysse and related network-management topics. The retrieved material was used as background for topic selection and citation drafting. No claim is made that the tool automatically established a complete citation graph containing every paper cited by, or citing, every document in the knowledge base.

G. Human Verification Required

Before submission, the author should independently verify:

- the exact CNSM 2026 template and page limit;
- the final compiled page count;
- every bibliographic entry, DOI, page range, and publication venue;
- whether workshop and repository publications are acceptable under the conference rules;
- the accuracy of all descriptions of previous work;
- the originality of the proposed contribution;
- that no statement is interpreted as an experimental result unless supported by an actual experiment.